

FactoDual-X: A Dual-Branch Transformer with Cross-Modal Attention for Hyperspectral Image Classification



Jalaj Gupta

Advisor: **Dr. Kaustuv Nag**

Department of Computer Science and Engineering
Indian Institute of Information Technology Guwahati

This dissertation is submitted for the degree of
Master of Technology

to my loving parents...

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others except as specified in the text and Acknowledgements.

Jalaj Gupta

Roll: 2521513,

Department of Computer Science and Engineering,
Indian Institute of Information Technology Guwahati.

Certificate

This is to certify that the thesis entitled “**Hyperspectral Image Classification using FactoDual-X Transformer**”, submitted by **Jalaj Gupta** to the Indian Institute of Information Technology Guwahati in partial fulfillment of the requirements for the award of the degree of **Master of Technology** in Computer Science and Engineering, is a record of original research work carried out by him under my supervision and guidance.

To the best of my knowledge, this thesis has not been submitted, either in part or in full, for the award of any degree, diploma, or any other academic qualification at any institution.

Dr. Kaustuv Nag

Assistant Professor,

Department of Computer Science and Engineering,

Indian Institute of Information Technology Guwahati.

Acknowledgements

This thesis would not have been possible without the constant support, guidance, and a healthy amount of confusion, caffeine, and last-minute panic from the people around me.

First and foremost, I would like to sincerely thank Dr. Kaustuv Nag for his valuable guidance, patience, and continuous support throughout this project. His insights and encouragement played a crucial role in shaping this work, especially when things were not working (which, to be honest, was quite often). I am also grateful for the occasional “motivational pressure” that kept me on track — and I still believe I am owed at least one treat for successfully completing this thesis.

A special thanks to my friends — for the discussions that sometimes helped, sometimes confused, but always made the journey memorable. From debugging code at odd hours, questioning life decisions during training runs, to celebrating small victories like a working model (or even a running code), they made this entire experience much more enjoyable. Without them, this journey would have been far less fun and far more stressful.

Finally, and most importantly, I would like to thank my parents and family. Their constant support, encouragement, and belief in me kept me going even during the most challenging times. Even when they didn’t fully understand what I was doing, they never stopped supporting me — and that means everything. This work is as much theirs as it is mine.

Abstract

This thesis addresses the challenge of accurate classification of hyperspectral images (HSI) by proposing a transformer-based approach that effectively captures both spectral and spatial information. Hyperspectral images contain rich spectral signatures for each pixel, but their high dimensionality and limited labeled data make classification a challenging task.

We begin by examining existing hyperspectral classification methods, particularly those that rely on convolutional neural networks and spectral-only transformer models. While these methods have shown promising results, they often fail to fully exploit the joint spectral-spatial dependencies present in hyperspectral data. In particular, spectral-only approaches underutilize spatial context, whereas purely spatial models fail to capture the detailed spectral characteristics required for accurate classification.

To address these limitations, we propose a dual-branch transformer-based architecture, referred to as FactoDual-X. The proposed model introduces a dual tokenization strategy, where hyperspectral data is decomposed into spatial tokens and spectral tokens. Spatial tokens capture local neighborhood relationships, while spectral tokens represent grouped spectral band information.

A key component of the proposed approach is the use of cross-modal attention, which enables effective interaction between spatial and spectral representations. In this mechanism, spatial features attend to spectral features, allowing the model to refine its predictions by integrating both structural and material information.

Furthermore, the model employs transformer encoders to learn long-range dependencies across both domains, followed by feature aggregation and classification using a fully connected layer. To handle class imbalance in the dataset, a weighted cross-entropy loss function is utilized during training.

The proposed method is implemented and evaluated on the Indian Pines dataset. Experimental results demonstrate that the model achieves high classification accuracy, validating the effectiveness of combining spectral-spatial modeling with attention-based feature fusion.

Table of Contents

List of Figures	ix
1 Introduction	1
1.1 Motivation	1
1.2 Objective of the Thesis	2
1.3 Contribution of the Thesis	2
2 Related Works	3
3 Proposed Model	5
3.1 Overview of the Proposed Model	5
3.2 Data Representation and 3D Patch Embedding	5
3.3 Parallel Factorized Encoding	6
3.3.1 Spatial Branch	6
3.3.2 Spectral Branch	7
3.4 Cross-Modal Attention Fusion	7
3.5 Feature Integration and Output	8
3.6 Computational Efficiency	8
3.7 Summary of the Proposed Approach	9
4 Experimentation	10
4.1 Experimental Setup	10
4.2 Data Preprocessing	10
4.3 Training-Test Split Statistics	11
4.4 Implementation Details	11
4.5 Evaluation Metrics	12
4.6 Experimental Results	12
4.7 Per-Class Performance	13
4.8 Analysis	13

Table of Contents

4.9 Visualization Results 14

5 Conclusions, Limitations, and Future Plans 16

5.1 Conclusions and Limitations 16

5.2 Future Plans 17

References 18

List of Figures

4.1	Visual comparison of classification maps on the Indian Pines dataset. The top row shows Ground Truth, SpectralFormer, MAEST, and FactorFormer, while the bottom image shows the proposed FactoDual-X model. The proposed method produces cleaner and more consistent class regions.	14
4.2	Training loss and accuracy curves of FactoDual-X.	15

Chapter 1

Introduction

Hyperspectral image (HSI) classification is an important problem in remote sensing, environmental monitoring, and agriculture. Unlike RGB images, HSIs contain hundreds of spectral bands per pixel, providing rich material information. However, this high dimensionality leads to challenges such as spectral redundancy, limited labeled data, and difficulty in modeling joint spatial-spectral relationships.

Traditional methods such as SVMs and CNNs have shown good performance but are limited in capturing long-range dependencies and often fail to fully utilize spectral information. Similarly, many transformer-based approaches focus mainly on spectral modeling, neglecting spatial context.

To address these issues, this thesis proposes a transformer-based approach that jointly models spatial and spectral information using dual-token representation and attention-based feature fusion, enabling more effective hyperspectral classification.

1.1 Motivation

Recent advancements in transformer architectures have demonstrated their effectiveness in modeling long-range dependencies in various domains, including natural language processing and computer vision. In hyperspectral image classification, transformers provide a powerful alternative to convolution-based models due to their ability to capture global contextual relationships.

However, existing hyperspectral transformer models exhibit several limitations. Many approaches rely primarily on spectral tokenization, leading to insufficient utilization of spatial information. Spatial correlations between neighboring pixels are

not effectively modeled, and the limited availability of labeled data in hyperspectral datasets makes training deep models challenging.

Additionally, hyperspectral data contains strong interdependence between spectral signatures and spatial structures. Ignoring either aspect leads to suboptimal classification performance. Therefore, there is a need for a model that can effectively integrate both spatial and spectral features in a unified framework.

1.2 Objective of the Thesis

The main objective of this thesis is to develop an effective transformer-based framework for hyperspectral image classification that addresses the limitations of existing methods.

Specifically, the objectives of this work are to design a model that captures both spectral and spatial dependencies simultaneously, to introduce a dual-token representation for efficient modeling of hyperspectral data, to incorporate cross-modal attention for effective fusion of spatial and spectral features, and to improve classification accuracy on benchmark hyperspectral datasets.

1.3 Contribution of the Thesis

This thesis presents a transformer-based architecture, referred to as FactoDual-X, for hyperspectral image classification. The key contributions of this work are summarized as follows.

A dual-token representation is introduced, where hyperspectral data is decomposed into spatial tokens and spectral tokens, enabling separate modeling of spatial and spectral information. A transformer-based architecture is designed to learn long-range dependencies across both domains using self-attention mechanisms. A cross-modal attention mechanism is incorporated to enable effective interaction between spatial and spectral representations, improving feature fusion. A patch-based input strategy is employed to capture local spatial context while maintaining computational efficiency. Extensive experiments are conducted on the Indian Pines dataset, demonstrating significant improvement in classification performance, achieving high accuracy compared to baseline approaches.

Chapter 2

Related Works

The development of transformer-based architectures has significantly transformed the field of machine learning, particularly in sequence modeling and representation learning. The foundation of this progress can be traced back to the introduction of the Transformer architecture in the seminal work “Attention Is All You Need” [1], where self-attention mechanisms were proposed as an alternative to recurrent and convolutional operations. By enabling direct modeling of global dependencies within input sequences, transformers provided improved parallelization and superior performance across a wide range of tasks.

Building upon this foundation, subsequent research focused on improving representation learning capabilities. One of the most influential models in this direction is BERT (Bidirectional Encoder Representations from Transformers) [2], which introduced bidirectional pretraining using masked language modeling. Unlike earlier unidirectional models, BERT captures contextual information from both left and right directions, leading to substantial improvements in various natural language understanding tasks.

Despite their success, standard transformer models are limited by fixed-length context windows, restricting their ability to model long-range dependencies. To address this limitation, Transformer-XL [3] was proposed, introducing a segment-level recurrence mechanism that allows information to be propagated across segments. This approach effectively extends the context length and improves the modeling of long-term dependencies without increasing computational cost significantly.

As transformer architectures scaled to larger inputs, computational efficiency became a critical challenge due to the quadratic complexity of self-attention. To mitigate

this issue, several efficient transformer variants were introduced. The Reformer [4] employs locality-sensitive hashing to approximate attention computation, reducing complexity from quadratic to logarithmic order. Similarly, the Longformer [5] introduces a combination of local windowed attention and global attention mechanisms, enabling linear scalability with respect to sequence length and making it suitable for long document processing.

In parallel, transformer models were also extended to unified learning paradigms. The Text-to-Text Transfer Transformer (T5) framework [6] reformulates all natural language processing tasks into a unified text-to-text format, allowing a single model to handle diverse tasks through a consistent architecture and objective function. This unification further demonstrated the flexibility and scalability of transformer-based approaches.

While these advancements have primarily focused on natural language processing, transformers have also been successfully adapted to computer vision and hyperspectral image analysis. In hyperspectral image (HSI) classification, the high dimensionality and complex spectral-spatial relationships pose unique challenges. Traditional methods, including CNN-based approaches, struggle to capture long-range dependencies across spectral bands.

Recent transformer-based approaches have shown promise in addressing these limitations by modeling global dependencies. However, many existing models focus primarily on spectral tokenization and fail to fully exploit spatial information present in hyperspectral data.

To overcome these challenges, the FactoFormer framework [7] introduces a factorized transformer architecture that separately models spectral and spatial dimensions. By decomposing the attention mechanism into two parallel branches, FactoFormer effectively captures intra-domain relationships while reducing computational complexity. Furthermore, the framework incorporates self-supervised pretraining strategies to address the scarcity of labeled hyperspectral data.

Motivated by these advancements, this thesis builds upon the principles of factorized transformer architectures and proposes an enhanced dual-token representation model, FactoDual-X. The proposed approach extends the idea of spectral-spatial factorization by introducing cross-modal attention mechanisms, enabling more effective interaction between spatial and spectral features and resulting in improved hyperspectral image classification performance.

Chapter 3

Proposed Model

3.1 Overview of the Proposed Model

In this work, we propose a transformer-based architecture, termed FactoDual-X, for hyperspectral image (HSI) classification. The model is designed to effectively capture the complex spatio-spectral characteristics of hyperspectral data through a dual-branch factorized transformer architecture, inspired by recent advances in factorized transformer models [7].

Unlike conventional approaches, the proposed model decomposes the learning process into two parallel streams that independently model spatial and spectral dependencies. These representations are subsequently fused using a cross-modal attention mechanism, enabling the model to learn a unified and discriminative feature representation. This design improves classification performance while maintaining computational efficiency.

3.2 Data Representation and 3D Patch Embedding

Let the hyperspectral image be represented as: $X \in \mathbb{R}^{H \times W \times C}$

where H and W denote the spatial dimensions and C represents the number of spectral bands.

To preserve local spatial-spectral context, the input is partitioned into 3D patches:

$$x_p \in \mathbb{R}^{P \times P \times B}$$

where x_p is partitioned set of D patches, patches P is the spatial patch size and B is the spectral grouping interval.

Each patch is flattened and projected into a D -dimensional embedding space using a learnable linear projection:

$$Z_0 = [x_p^1 E; x_p^2 E; \dots; x_p^N E] + E_{pos}$$

where Z_0 is the initial token representation (embedding) that is fed into the transformer and each patch x_p^i is first flattened into a vector of dimension $P \cdot P \cdot C$ before being projected into a D -dimensional embedding space using the learnable projection matrix E and E_{pos} represents positional encoding it works as a coordinate.

3.3 Parallel Factorized Encoding

To reduce computational complexity and improve feature learning, the transformer is factorized into two parallel branches [7]: a spatial Branch (B_{spa}) and a spectral Branch (B_{spe}).

3.3.1 Spatial Branch

The spatial branch captures morphological relationships among neighboring pixels. Given spatial tokens $Z_s \in \mathbb{R}^{N_{spa} \times D}$, the self-attention mechanism is defined as:

$$Q_s = Z_s W_Q, \tag{3.1}$$

$$K_s = Z_s W_K, \tag{3.2}$$

$$V_s = Z_s W_V \tag{3.3}$$

$$\text{SSA}(Q_s, K_s, V_s) = \text{Softmax} \left(\frac{Q_s K_s^T}{\sqrt{d_k}} \right) V_s \tag{3.4}$$

where Z_s denotes the spatial token embeddings, with N_{spa} representing the number of spatial tokens and D the embedding dimension. The matrices $Q_s, K_s, V_s \in$

$\mathbb{R}^{N_{spa} \times d_k}$ correspond to the query, key, and value representations, respectively. The learnable projection matrices $W_Q^s, W_K^s, W_V^s \in \mathbb{R}^{D \times d_k}$ map the input embeddings into the query, key, and value spaces, where d_k denotes the dimension of each attention head.

The operation $SSA(\cdot)$ represents the spatial self-attention mechanism, where attention weights are computed using scaled dot-product attention, and the Softmax function ensures proper normalization of attention scores.

3.3.2 Spectral Branch

The spectral branch models inter-band dependencies across spectral channels. Given spectral tokens

$$Q_c = Z_c W_Q^c, \quad (3.5)$$

$$K_c = Z_c W_K^c, \quad (3.6)$$

$$V_c = Z_c W_V^c, \quad (3.7)$$

$$SpeSA(Q_c, K_c, V_c) = \text{Softmax} \left(\frac{Q_c K_c^T}{\sqrt{D}} \right) V_c \quad (3.8)$$

where Z_c denotes the spectral token embeddings, with N_{spe} representing the number of spectral tokens and D the embedding dimension. The matrices $Q_c, K_c, V_c \in \mathbb{R}^{N_{spe} \times D}$ correspond to the query, key, and value representations in the spectral domain. The learnable projection matrices $W_Q^c, W_K^c, W_V^c \in \mathbb{R}^{D \times D}$ map the input embeddings into the respective attention spaces.

The function $SpeSA(\cdot)$ represents the spectral self-attention mechanism, which captures inter-band dependencies by computing attention weights across spectral tokens.

3.4 Cross-Modal Attention Fusion

To integrate spatial and spectral representations, a cross-modal attention mechanism is introduced. In this process, spatial tokens act as queries, while spectral tokens

serve as keys and values, extending the factorized attention concept for effective feature fusion [7].

$$M = \text{Softmax} \left(\frac{Q_s(\Phi K_c)^T}{\sqrt{d_{\text{head}}}} \right) \quad (3.9)$$

$$F_{\text{fusion}} = M(\Phi V_c) \quad (3.10)$$

The cross-modal attention computes an attention map M using spatial queries Q_s and projected spectral keys ΦK_c , where d_{head} is the scaling dimension. The fused output F_{fusion} is obtained by weighting the projected spectral values ΦV_c using M .

Φ denotes a linear projection used for dimensional alignment.

This mechanism allows spatial features to selectively attend to relevant spectral information, resulting in a more discriminative and robust representation.

3.5 Feature Integration and Output

The fused representation is passed through normalization and a feed-forward network:

$$Z_{\text{out}} = \text{MLP}(\text{LN}(F_{\text{fusion}} + Z_s)) + (F_{\text{fusion}} + Z_s)$$

This residual formulation improves training stability and convergence. The final output Z_{out} is obtained by applying layer normalization (LN) and a feed-forward network (MLP) to the fused features $(F_{\text{fusion}} + Z_s)$, followed by a residual connection. This formulation improves feature representation while ensuring stable training and efficient gradient flow.

3.6 Computational Efficiency

The proposed factorized architecture reduces computational complexity from:

$$O(N^2) \rightarrow O(N_{\text{spa}}^2 + N_{\text{spe}}^2)$$

This makes the model scalable for high-dimensional hyperspectral data while maintaining strong performance. The computational complexity is reduced from $O(N^2)$ to $O(N_{\text{spa}}^2 + N_{\text{spe}}^2)$, where N is the total number of tokens, and N_{spa} and N_{spe} denote spatial and spectral tokens, respectively.

3.7 Summary of the Proposed Approach

The proposed FactoDual-X model separates spatial and spectral feature learning into parallel branches, utilizes transformer-based attention to capture long-range dependencies, integrates features using cross-modal attention, and achieves efficient computation with improved classification accuracy.

Chapter 4

Experimentation

4.1 Experimental Setup

To evaluate the effectiveness of the proposed FactoDual-X model, experiments are conducted on the Indian Pines hyperspectral dataset. The dataset consists of 145×145 pixels with 200 spectral bands and 16 land-cover classes.

Due to the inherent class imbalance in the dataset, a stratified sampling strategy is adopted. Specifically, approximately 10% of samples from each class, with a minimum threshold per class, are used for training, while the remaining samples are used for testing. This ensures fair representation of all classes during training.

4.2 Data Preprocessing

Prior to training, the hyperspectral data is normalized using per-band z-score normalization. This normalization ensures zero mean and unit variance across spectral bands, which improves convergence and stabilizes training.

To incorporate spatial context, a patch-based input representation is employed. For each pixel, a local spatial neighborhood of size 11×11 is extracted along with all 200 spectral bands, resulting in an input patch of size $11 \times 11 \times 200$. This enables the model to jointly learn spatial and spectral features.

4.3 Training-Test Split Statistics

The distribution of samples across training and testing sets for each class is presented in Table 4.1. A stratified sampling strategy is employed, wherein approximately 10% of samples from each class are used for training and the remaining 90% for testing, ensuring balanced class representation despite dataset imbalance.

Class	Total	Train	Test
Alfalfa	46	15	31
Corn-notill	1428	142	1286
Corn-mintill	830	83	747
Corn	237	23	214
Grass-pasture	483	48	435
Grass-trees	730	73	657
Grass-pasture-mowed	28	14	14
Hay-windrowed	478	47	431
Oats	20	10	10
Soybean-notill	972	97	875
Soybean-mintill	2455	245	2210
Soybean-clean	593	59	534
Wheat	205	20	185
Woods	1265	126	1139
Buildings-Grass-Trees-Drives	386	38	348
Stone-Steel-Towers	93	15	78
Total	10249	1055	9194

Table 4.1 Per-Class Training and Testing Split on Indian Pines Dataset

4.4 Implementation Details

The proposed FactoDual-X model is implemented using PyTorch. The architecture follows a dual-branch transformer design consisting of separate spatial and spectral token processing modules.

The model uses an embedding dimension of 64 with four transformer layers and four attention heads. Training is performed with a batch size of 32 using the AdamW optimizer and an initial learning rate of 3×10^{-4} .

To improve convergence, a warmup strategy is applied during the initial training phase, followed by cosine decay scheduling. Additionally, weighted cross-entropy loss is employed to handle class imbalance effectively.

The total number of model parameters is approximately 432,848, and the architecture processes 121 spatial tokens and 20 spectral tokens per input sample.

4.5 Evaluation Metrics

The performance of the proposed model is evaluated using standard hyperspectral classification metrics, including Overall Accuracy (OA), Average Accuracy (AA), and Cohen’s Kappa coefficient (κ). These metrics provide a comprehensive evaluation of both global performance and class-wise prediction consistency.

4.6 Experimental Results

The training process is carried out for 300 epochs using a GPU-based setup. The model shows steady convergence, with training accuracy gradually improving and validation accuracy stabilizing in later epochs.

The performance is evaluated using Overall Accuracy (OA), Average Accuracy (AA), and the Kappa coefficient (κ). OA measures the proportion of correctly classified samples over the entire dataset, while AA computes the average of per-class accuracies, ensuring balanced evaluation across all classes. The Kappa coefficient assesses the agreement between predicted and true labels while accounting for random chance. The model achieves an OA of 95.99%, AA of 96.54%, and κ of 0.9541, indicating strong classification performance and generalization capability.

The final evaluation results on the Indian Pines dataset are summarized in Table 4.2.

Metric	Value
Overall Accuracy (OA)	95.99%
Average Accuracy (AA)	96.54%
Kappa Coefficient (κ)	0.9541

Table 4.2 Performance of FactoDual-X on Indian Pines Dataset

Model	OA (%)	AA (%)	κ
SpectralFormer	81.76	87.81	0.7919
MAEST	84.15	90.97	0.8200
FactoFormer	91.30	94.30	0.9006
FactoDual-X (Ours)	95.99	96.54	0.9541

Table 4.3 Comparison with State-of-the-Art Methods on Indian Pines Dataset

Class	Precision	Recall	F1-score
Alfalfa	0.8378	1.0000	0.9118
Corn-notill	0.9561	0.9479	0.9520
Corn-mintill	0.9676	0.9197	0.9430
Corn	0.9682	0.9953	0.9816
Grass-pasture	0.9975	0.9287	0.9619
Grass-trees	0.9954	0.9970	0.9962
Grass-pasture-mowed	0.6190	0.9286	0.7429
Hay-windrowed	0.9977	0.9884	0.9930
Oats	0.8333	1.0000	0.9091
Soybean-notill	0.9512	0.9131	0.9318
Soybean-mintill	0.9393	0.9810	0.9597
Soybean-clean	0.9182	0.8614	0.8889
Wheat	0.9343	1.0000	0.9661
Woods	0.9956	0.9939	0.9947
Buildings-Grass-Trees-Drives	0.9637	0.9914	0.9773
Stone-Steel-Towers	0.8966	1.0000	0.9455

Table 4.4 Per-Class Classification Performance of FactoDual-X

4.7 Per-Class Performance

To further analyze the model performance, class-wise precision, recall, and F1-score are evaluated. The results are presented in Table 4.4.

4.8 Analysis

The results indicate that the proposed FactoDual-X model achieves high classification accuracy by effectively capturing both spatial and spectral dependencies. The dual-token representation enables independent learning of spatial and spectral features, while the cross-modal attention mechanism facilitates efficient feature fusion.

The model demonstrates near-perfect classification performance for several classes such as Hay-windrowed, Woods, and Wheat. Slightly lower performance is observed in classes with limited training samples, such as Grass-pasture-mowed, which is expected due to data imbalance.

The use of transformer-based attention allows the model to capture long-range dependencies, while the warmup and cosine decay learning strategy contributes to stable convergence.

4.9 Visualization Results

Furthermore, a qualitative comparison of classification maps is presented in Fig. 4.1. It can be observed that the proposed FactoDual-X model produces significantly cleaner and more homogeneous regions compared to existing transformer-based methods.

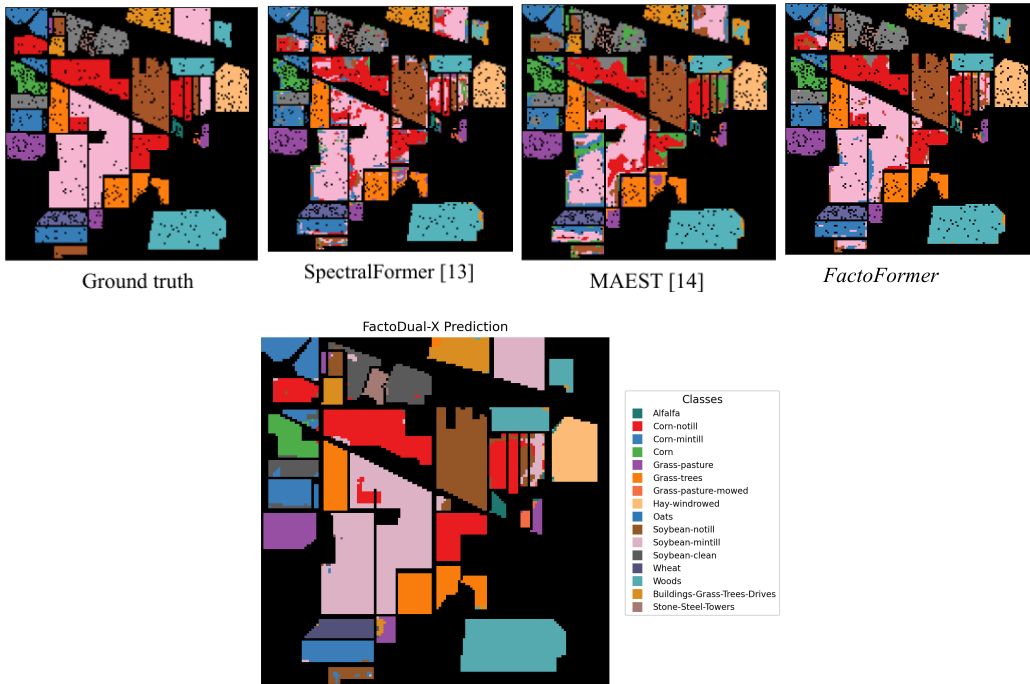


Fig. 4.1 Visual comparison of classification maps on the Indian Pine dataset. The top row shows Ground Truth, SpectralFormer, MAEST, and FactoFormer, while the bottom image shows the proposed FactoDual-X model. The proposed method produces cleaner and more consistent class regions.

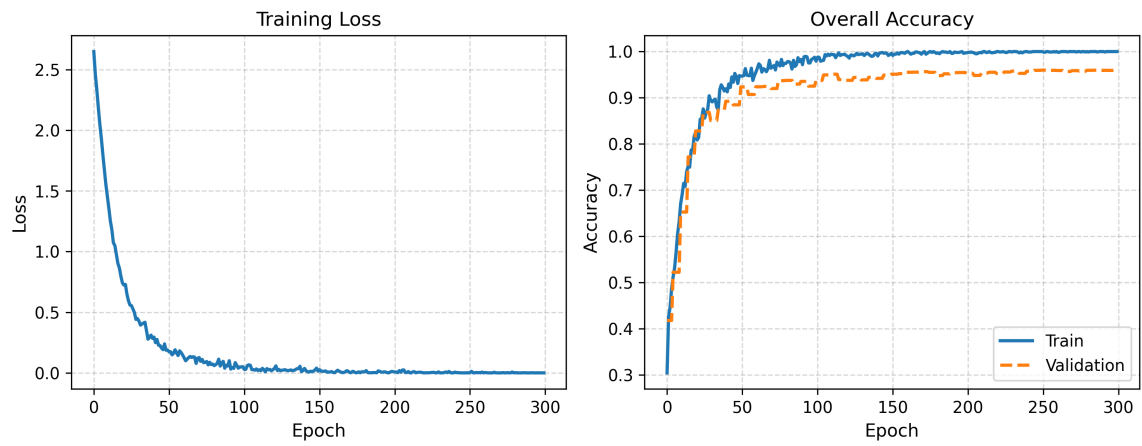


Fig. 4.2 Training loss and accuracy curves of FactoDual-X.

The training curves show smooth convergence, confirming stable optimization throughout the training process.

Chapter 5

Conclusions, Limitations, and Future Plans

5.1 Conclusions and Limitations

In this thesis, we proposed a transformer-based architecture, termed **FactoDual-X**, for hyperspectral image (HSI) classification. The model is designed to effectively capture both spectral and spatial dependencies using a dual-branch factorized transformer framework. By introducing a dual-token representation and integrating features through cross-modal attention, the proposed method successfully models the complex characteristics of hyperspectral data.

The experimental results on the Indian Pines dataset demonstrate that the proposed model achieves high classification performance, with an overall accuracy of 95.99%, validating the effectiveness of combining spatial and spectral information within a unified attention-based framework. The use of transformer-based architectures enables the model to capture long-range dependencies, while the factorized design improves computational efficiency.

Despite these promising results, the proposed approach has certain limitations. The model requires significant computational resources due to the use of transformer layers and attention mechanisms. Additionally, hyperspectral datasets often suffer from limited labeled samples, which can affect generalization performance. The training process is also time-consuming due to the large number of tokens generated from patch-based inputs.

5.2 Future Plans

Future work can focus on improving the scalability and efficiency of the proposed model by exploring lightweight transformer architectures and efficient attention mechanisms. Performance can be further enhanced through advanced data augmentation and semi-supervised or self-supervised learning to address limited labeled data. Extending the model to larger and more diverse hyperspectral datasets may improve robustness and generalization. Additionally, hybrid architectures combining convolutional and transformer-based methods can be investigated, along with more comprehensive evaluations against state-of-the-art approaches.

References

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017. 3
- [2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pp. 4171–4186, 2019. 3
- [3] Z. Dai, Z. Yang, Y. Yang, J. G. Carbonell, Q. Le, and R. Salakhutdinov, "Transformer-xl: Attentive language models beyond a fixed-length context," in *Proceedings of the 57th annual meeting of the association for computational linguistics*, pp. 2978–2988, 2019. 3
- [4] N. Kitaev, Ł. Kaiser, and A. Levskaya, "Reformer: The efficient transformer," *arXiv preprint arXiv:2001.04451*, 2020. 4
- [5] I. Beltagy, M. E. Peters, and A. Cohan, "Longformer: The long-document transformer," *arXiv preprint arXiv:2004.05150*, 2020. 4
- [6] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," *Journal of machine learning research*, vol. 21, no. 140, pp. 1–67, 2020. 4
- [7] S. Mohamed, M. Haghghat, T. Fernando, S. Sridharan, C. Fookes, and P. Moghadam, "Factoformer: Factorized hyperspectral transformers with self-supervised pretraining," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–14, 2024. 4, 5, 6, 8